

AI-Powered Student Success Analytics – Product Requirements Document (PRD)

1. Overview

This Product Requirements Document (PRD) consolidates institutional challenges, goals, and workflows from Bishop State, the University of Akron, and KCTCS, aligning them with the AI-Powered Student Success Analytics platform.

The purpose of this document is to define clear product requirements, success criteria, user roles, data workflows, and deliverables for the CodeBenders Datathon solution.

2. Problem Statement

Across all three institutions, student success initiatives suffer from inconsistent data access, siloed data systems, limited predictive capabilities, and slow reporting workflows. Institutions lack unified readiness metrics, actionable dashboards, and scalable predictive analytics needed to improve retention, course success, and credential completion.

3. Primary Users

- Advisors – Need early-warning insights and student-level risk indicators.
- Institutional Researchers – Need structured access to PDP + AR files + SIS data for analysis.
- Faculty – Need course-level success indicators, gateway course insights, and readiness trends.
- Leadership – Needs high-level retention, readiness, and enrollment metrics for resource planning.
- IT/Data Teams – Need a streamlined, automated, validated data submission and ingestion workflow.

4. Goals & Objectives

1. Deliver a unified analytics dashboard that integrates PDP, AR, and institutional data.
2. Provide predictive models for retention, gateway course success, readiness, and credential outcomes.
3. Enable natural-language queries for fast, self-service analytics.
4. Reduce institutional bottlenecks in data access, submission, and usage.
5. Improve student success metrics by enabling early, data-informed interventions.

5. Scope

IN SCOPE:

- Data ingestion pipeline (PDP → AR merge → institutional sources → warehouse).
- Unified dashboard with NLQ (natural-language querying).

- Core predictive models (retention, at-risk early warning, gateway course success, GPA prediction).
- Readiness index calculation (0–100%).
- Exportable visualizations and role-based access.

OUT OF SCOPE (for Datathon):

- Real-time pipelines beyond PDP/AR files.
- SIS system integration requiring institutional credentials.
- Long-term production support or hosting.

6. Institutional Requirements

KCTCS Requirements:

- Weekly readiness metrics included in enrollment workflows.
- Improved PDP adoption and clearer insight extraction.
- Automated PDP validation prior to submission.
- Dashboard showing: readiness, gateway course preparedness, underserved student flagging.

Bishop State Requirements:

- Improved access to PDP dashboards (role-based access).
- A faculty-facing AI tool for chart generation and querying.
- Course sequencing and combination insights.
- Data request workflow documentation and automation.
- Identification of high-risk gateway courses.

University of Akron Requirements:

- Predictive modeling based on 8 years of PDP submissions.
- Merge PDP + AR + local student data for baseline metrics.
- Improve adoption of dashboards among stakeholders.

- Gateway course success insights and readiness calculations.
- Identification of trends that correlate with improved retention.

7. Functional Requirements

FR1. Data Integration

- System must ingest PDP cohort and course files.
- System must ingest AR files and merge with PDP using unique student IDs.
- System must allow mapping to institutional data schemas.

FR2. Readiness Assessment

- System must compute readiness score (0–100%).
- Must support analysis of academic placement, course combinations, and preparedness.

FR3. Predictive Analytics

- Six predictive models: retention, at-risk, GPA prediction, gateway success, time-to-credential, credential type.
- Must provide insight explanations and feature importance.

FR4. Dashboard Requirements

- Must support customizable widgets.
- Must allow filtering by term, demographic attributes, courses, and credential types.
- Must support export to PDF/PNG.

FR5. AI Querying

- NLQ interface must allow users to ask questions about:
 - Retention trends
 - Readiness distributions
 - Gateway course performance
 - Demographic equity gaps

FR6. Role-Based Access

- Admin, Advisor, IR, Faculty, Leadership roles.
- Access rules must define PDP visibility, AR visibility, and student-level data controls.

FR7. Reporting

- Weekly, quarterly, and annual reports must be exportable.

8. Non-Functional Requirements

NFR1. Performance – Dashboard responses must render within 2–4 seconds for typical queries.

NFR2. Security – PDP and AR files contain PII; encryption at rest + access control required.

NFR3. Maintainability – Models must be retrainable as new cohorts are added.

NFR4. Usability – Dashboards must be accessible to non-technical users.

NFR5. Auditability – Data transformations must be traceable for compliance (federal/state reporting).

9. Data Pipeline Requirements

- Must clean, validate, and conform PDP files to required schema.
- Must include PDP pre-submission validation tool.
- Must support merging PDP cohort + PDP course + AR files.
- Must make resulting datasets queryable via dashboard and NLQ service.
- Data must be refreshed quarterly and optionally weekly where feasible.

10. Success Metrics

- Reduction in time required to generate enrollment and readiness reports (target: 60%).
- Increased adoption of dashboards by advisors and leadership (target: 3× increase).
- Improved ability to identify underprepared students before term start.
- Increase in predictive model accuracy over baseline institutional methods.
- Demonstrated retention insights used in resource planning or grants.

11. Risks & Assumptions

RISKS:

- AR and PDP data schemas vary by institution.
- Single-person data silos may introduce delays.
- Annual submission cycles limit real-time insights.

ASSUMPTIONS:

- PDP + AR data is accessible and provided during Datathon.
- Institutions will share representative sample data.
- Dashboard usage patterns will mirror reported workflows.

12. Next Steps

- Finalize institutional mappings and problem statements.
- Build unified PDP+AR ingestion scripts.
- Implement readiness index and two priority predictive models.
- Develop NLQ prototype and core dashboard widgets.
- Deliver cross-institution insights summary.